

Squeeze casting: an overview

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Abstract

This paper reviews squeeze casting in all its aspects: its origins and developments; the various processes and equipment involved including the major parameters; metallurgical features including porosity, recrystallisation and grain refinement; mechanical properties of the products; the mechanics of the different processes, and the advantages and disadvantages of squeeze casting. © 2000 Elsevier Science S.A. All rights reserved.

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1. Introduction

Casting is the most economical route to transfer raw materials into readily usable components. However, one of the major drawbacks for conventional or even more advanced casting techniques, e.g., high pressure die-casting [1–5] is the formation of defects such as porosity. Furthermore, segregation defects of hot tears, A and V segregates and banding [6–12] could be potential crack initiators during service operation of the as-cast components. New casting techniques have, therefore, been developed to compensate for these shortcomings. Of the many such casting techniques available, squeeze casting has greater potential to create less defective cast components.

Squeeze casting (SC) is a generic term to specify a fabrication technique where solidification is promoted under high pressure within a re-usable die. It is a metal-forming process, which combines permanent mould casting with die forging into a single operation where molten metal is solidified under applied hydrostatic pressure. Although squeeze casting is now the accepted term for this forming operation, it has been variously referred to as “extrusion casting” [13], “liquid pressing” [14], “pressure crystallisation” [15] and “squeeze forming” [16]. The idea was initially suggested by Chernov [17] in 1878 to apply steam

pressure to molten metal while being solidified. However, in spite of its century old invention, commercialisation of squeeze casting has been achieved only quite recently and is mainly concentrated in Europe and Japan. It is mainly used to fabricate high integrity engineering components with or without reinforcement [18]. Hartley [19] reported a technique developed by GKN Technology in UK for the pressurised solidification of Al alloy in reusable dies. In this process a die set is placed on a hydraulic press and preheated, and the exact amount of molten alloy is poured into the lower half of the open die set, the press closed so that the alloy fills the cavity and the pressure maintained until complete solidification occurs (31–108 MPa pressure). External undercut forms can be produced, and using retractable side cores, through-holes are possible. Since the as-fabricated components can be readily used in service or after a minor post-fabrication treatment, squeeze casting is also regarded as a net or near net-shape fabrication route.

Parallel to commercialisation, there are research centres throughout the world that are actively researching further development and exploitation of this net or near net-shape fabrication process. This is evidenced by the publication of more than 700 papers in various engineering and scientific journals. These are mainly related to aluminium and magnesium-based alloys with special emphasis on metal matrix composites MMCs. According to Crouch [20], squeeze casting is now the most popular fabrication route for MMC artefacts. The annual 12–15% growth rate of MMCs in the automotive, aerospace, sport and leisure

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goods and other markets [21] is a clear indication of better usage of advanced manufacturing routes such as squeeze casting.

Generally, the SC-fabricated engineering components are fine grained with excellent surface finish and have almost no porosity. They come in a variety of shapes and sizes. The mechanical properties of these parts are significantly improved over those of conventional castings and more sophisticated casting routes of pressure or gravity die-casting. According to Pennington [22], yield strength is improved by 10–15% and elongation and fatigue strength by as much as 50–80%. Dimensional accuracy is similar to those of die-casting; 0.25 mm in 100 mm to 0.6 mm in 500 mm. It is further claimed that SC-fabricated components have superior weldability and heat treatability. In addition, since squeeze casting may be carried out without any feeding system, runners, gates, etc., and shrinkage compensating units, risers, the yield is quite high with almost no scrap for recycling.

Finally, in contrast to forging, squeeze cast components are fabricated in a single action operation with lesser energy requirements.

2. Process outline

The process of squeeze casting involves the following steps:

- 1. A pre-specified amount of molten metal is poured into a preheated die cavity, located on the bed of a hydraulic press.
- 2. The press is activated to close off the die cavity and to pressurise the liquid metal. This is carried out very quickly, rendering solidification of the molten metal under pressure.
- 3. The pressure is held on the metal until complete solidification. This not only increases the rate of heat flow, but also most importantly may eliminate macro/micro shrinkage porosity. In addition, since nucleation of gas porosity is pressure-dependent [23], the porosity formation due to dissolve gases in the molten metal is restricted.

- 4. Finally the punch is withdrawn and the component is ejected.

3. Mechanics of squeeze casting

3.1. The die

A most crucial aspect in permanent mould castings such as die-casting or squeeze casting is the die itself and, most importantly, the design of the die including the selection of a suitable die material, the manufacturing process, appropriate heat treatment and the maintenance practice. Squeeze casting dies are exposed to severe thermal and mechanical cyclic loading, which may cause thermal fatigue, cracking, erosion, corrosion, and indentation. The nature and features of a die are greatly influenced by the particular alloy to be cast. Currently H13 tool steel is a widely used material of constructions but generally die steels should have good hot hardness, high temper resistance, adequate toughness and especially a high degree of cleanliness and uniform microstructure.

3.2. The casting process: key features

Two basic forms of the process may be distinguished, depending on whether the pressure is applied directly on to the solidifying cast product via an upper or male die (punch) or the applied pressure is exerted through an intermediate feeding system as schematically shown in Fig. 1: (i) the direct squeeze casting mode, and (ii) the indirect squeeze casting mode.

For the direct mode, two further forms may be distinguished based on liquid metal displacement initiated by the punch movement: (i) without metal movement, and (ii) with metal movement.

As illustrated in Fig. 2, the first form is suitable for ingot-type components where there is no metal movement, whilst the second type involving metal movement, also known as the backward process, is more versatile and can be used to cast a wide range of shaped components.

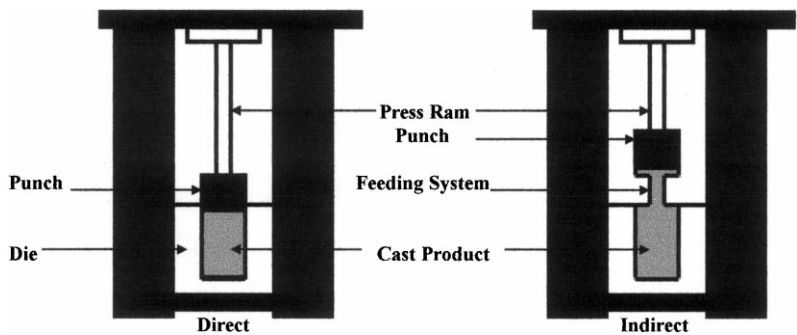


Fig. 1. Schematic diagram to illustrate the direct and indirect modes of the squeeze casting process.

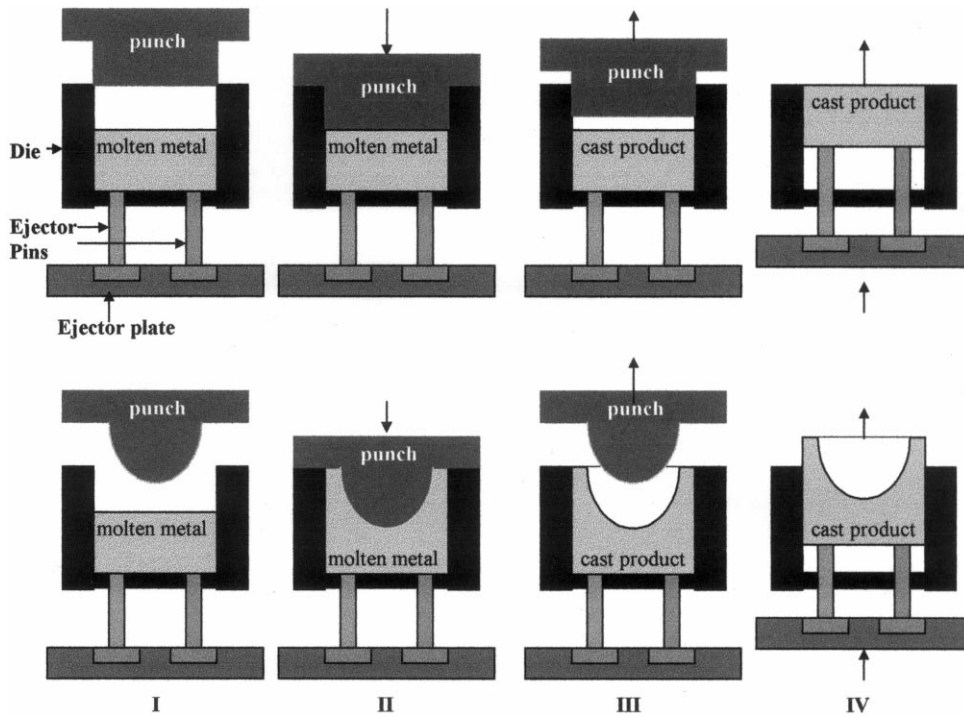


Fig. 2. Schematic diagram to show two forms of the direct squeeze casting process.

In addition to the molten metal displacement forms, the squeeze casting process may also be classified based on the following features.

3.2.1. Type of equipment

A variety of squeeze casting machines are in use in various parts of the world. They are either designed by the researchers themselves [24], the so-called home-made, or manufactured by machine tools companies on a mass production basis [25]. The direct SC-machines are simple and straight forward but the indirect ones generally fall into the following categories of: (i) vertical die closing and injection, (ii) horizontal die closing and injection, (iii) horizontal die closing and vertical injection, and (iv) vertical die closing and horizontal injection.

3.2.2. Timing of pressure application

Although squeeze casting is regarded as the pressurisation of molten alloy, it may also be used for shaping semi-solids and, therefore, a further classification may be envisaged as: (i) before the beginning of crystallisation, and (ii) after the beginning of crystallisation, which may also be described as semi-solid pressing.

Fig. 3 summarises the various modes of the squeeze casting process.

3.3. Process parameters

The most important process parameter is the alloy itself. The composition and physical characteristics of the alloy are of paramount importance due to their direct effects on the

die life. These include the melting temperature, and thermal conductivity of the alloy together with the combined effect of the heat-transfer coefficient and soldering onto the die material. Furthermore, the alloy dictates the selection of casting parameters such as die temperature, which has direct consequence on the die life. Therefore, squeeze casting is usually employed for low melting temperature alloys of aluminium and magnesium.

In addition to the composition of a casting alloy, which determines its freezing range and affects the quality of finished components, the casting parameters should also be controlled very closely to achieve a sound casting. The most dominant process parameters are die temperature and pouring temperature, and superheat, although the level of applied pressure is also important. Since the metal is cast under pressure, the inherent castability of the alloy is of little

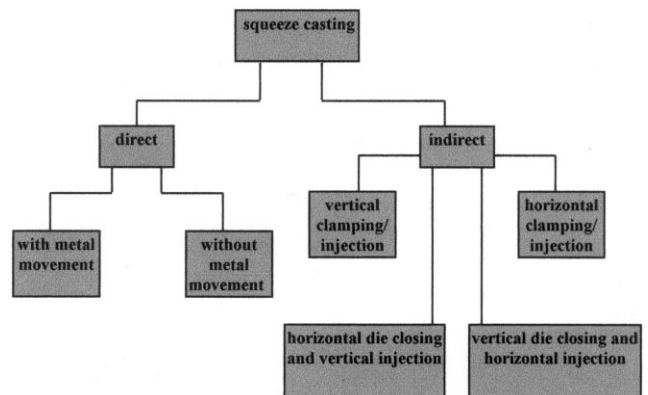


Fig. 3. Block diagram to show the squeeze casting classifications.

or no concern. Other important parameters include the cleanliness of the metal in relation to the presence of inclusions, metal movement within the die which may induce turbulence, the die coat, and the time interval over which the pressure is applied, i.e., the so-called dead time. The die temperature is usually held at between 200°C and 300°C for aluminium and magnesium alloys, whilst the applied pressure varies between 50 and 150 MPa. The lubrication medium, i.e., the die coat, is usually graphite based. Heat-transfer coefficients are extremely high due to the casting metal being pressed against the die wall.

4. Theoretical background: effect of pressure on the solidification behaviour of alloys

The application of pressure during solidification would be expected to affect phase relationships in an alloy system. This may be deduced by considering the Clausius–Clapeyron equation,

$$\frac{\Delta T_f}{\Delta P} = \frac{T_f(V_l - V_s)}{\Delta H_f}, \quad (1)$$

where T_f is the equilibrium freezing temperature, V_l and V_s are the specific volumes of the liquid and solid, respectively, and ΔH_f is the latent heat of fusion. Substituting the appropriate thermodynamic equation for volume,² the effect of pressure on freezing point may roughly be estimated as follows:

$$P = P_0 \exp\left(\frac{-\Delta H_f}{RT_f}\right), \quad (2)$$

where P_0 , ΔH_f and R are constants. Therefore, T_f should increase with increasing pressure. On a mechanistic approach, such change in freezing temperature is expected due to the reduction in interatomic distance with increasing pressure and thus restriction of atomic movement, which is the prerequisite for melting/freezing. The inter-solubility of constituent elements together with the solubility of impurity and trace elements is also expected to increase with pressure.

The above mentioned theoretical predictions have been proven experimentally where a liquidus temperature rise of up to 9°C has been reported for pure Al/Si binary alloys [26,27] at a pressure of ~150 MPa. Furthermore, the eutectic point moves to the left, i.e., to higher Si contents [28]. The consequences of such changes in the phase diagrams are a significant improvement in the microstructure and mechanical properties of SC-fabricated components. As reported by Chadwick and Yue [18] and Franklin et al. [29], grain refinement is quite noticeable in squeeze cast parts. However, their interpretation of such refinement is sharply different. Chadwick and Yue [18] have claimed that pressure has no effect on grain refinement, the observed fine

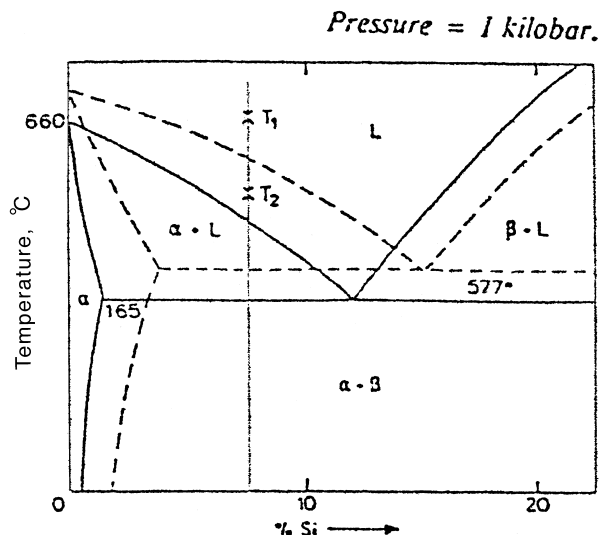


Fig. 4. The effect of rapid cooling and the application of pressure on the Al-Si phase diagram [28].

grained structure of squeeze castings being principally due to the increase in heat-transfer coefficients, i.e., greater cooling rates for the solidifying alloy due to reduction in the air gap between the alloy and the die wall and thus more effective contact area. Franklin et al. [29], however, have suggested the opposite, where the application of pressure brings about undercooling in an initially superheated alloy (see Fig. 4) and thus increases nucleation frequency so bringing about a finer grain size.

In the opinion of the present authors, both hypotheses are valid and are active simultaneously at a time but one may dominate the other at any instant during solidification. The size of the air gap between the solidifying alloy and the die wall and the degree of undercooling, the two main features for fine structure, are dependent on such process parameters as the pressure, the timing of its application and the chemistry of the solidifying metal. Certainly, the application of pressure reduces the air gap between the solidifying metal and the metallic mould and thus increases the contact area; effecting improvement in the heat-transfer coefficient, but this may not be the dominant mechanism. This is in line with studies carried out by Gethin et al. [30] who employed numerical modelling to simulate the flow and heat-transfer characteristics of molten metal during squeeze casting. They reported small changes in the thermo-physical properties of the molten metal with applied pressure.

The timing of pressure application is critical, since if it is applied at a temperature of $T > T_m + \Delta T$, ΔT being the expected increase in T_m due to pressure, the effect of undercooling is negligible and thus increase in the heat-transfer coefficient is the dominant mechanism. If however, the pressure is applied at a temperature $T_m \leq T \leq T_m + \Delta T$, then undercooling would be the dominant mechanism. The effect of undercooling is still appreciable if pressure is applied at the beginning of crystallisation, i.e., at a tem-

² The liquid metal is considered as an ideal gas.

perature $T < T_m$. The composition of the solidifying alloy is also quite important as, for instance, in case of aluminium alloys, the presence of some trace elements of Sr, P, Na or even Ti or B could have tremendous effects on the scale of the microstructure.

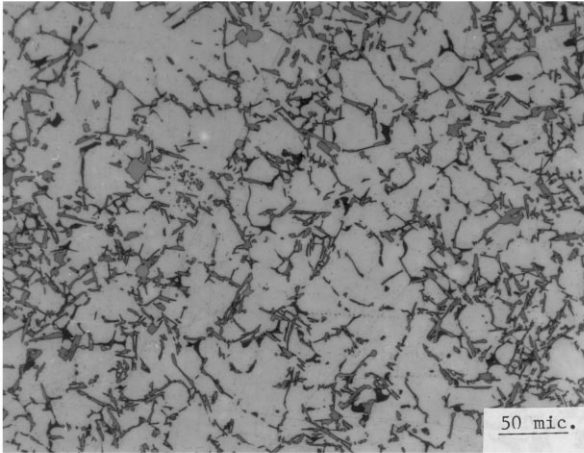
Irrespective of the mechanism responsible for grain refinement, investigation on Al–Li [31], Al–Li–Zr [31], Al–Li–Zr–Ti–B [31], Al–Li–Cu–Zr [31], Al–Cu [32], Al–Si [33] and pure Al [34] have shown improved mechanical properties of squeeze cast components with increasing pressure. For instance, Balan et al. [33] studied the effect of applied pressure (1–75 MPa) on LM6 Al–Si alloy and reported an increase of about 3.4% in density, 63% in UTS, 2.6% in percentage elongation and 50% in hardness with increasing applied pressure. Such remarkable improvement was mainly due to microstructural alteration through refinement of the primary α -phase and modification of the silicon phase. Similar results were obtained for pure aluminium [34].

In general, the fine structure and superior mechanical properties of SC-components are due to the following factors: (i) changes in undercooling of the molten alloy, (ii) changes in the composition and percentages of the forming phases of the solidifying alloy, (iii) changes in the heat-transfer coefficient between the metallic mould and solidifying alloy, and (iv) changes in the density of the alloy due to reduction of porosity.

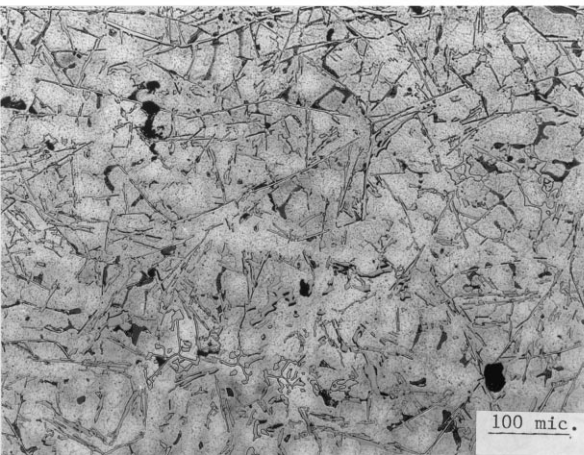
5. Squeeze cast alloys: microstructure and properties

As mentioned before, squeeze cast products have superior mechanical properties to their conventionally cast counterparts due to higher density, finer grain size and more homogenous microstructure [22,28,32,33,35]. Furthermore, thermal limitations of the die restricts squeeze casting to mainly light and low melting point alloys of aluminium and magnesium, although there are reports that copper alloys [36], cast irons [24,37] and steels [14] have also been squeeze cast. Fig. 5 summarises the alloys used for squeeze casting.

The application of pressure has a significant effect on the morphology of the phases in squeeze cast alloys as, for instance, there is a tendency for flake graphite to transform to compacted graphite with increasing pressure [37]. In the



(a)



(b)

Fig. 6. Optical micrographs to show the as-cast structure of LM24 Al–Si alloy [40]: (a) squeeze cast LM24, and (b) conventional as-cast LM24.

case of Al–Si alloys, fibrous silicon is the likely morphology. A typical squeeze cast microstructure of LM24 Al–Si alloys is shown in Fig. 6 together with its conventionally cast counterpart to highlight the differences in silicon morphology and grain structure. It is, therefore, expected that squeeze cast products will have superior mechanical properties, as illustrated in Table 1.

Some authors found it essential to design alloys specifically for SC conditions. New alloy system of Al–Si–Cu–Mg which posses good castability and excellent mechanical properties was examined by Lee et al [42] to optimise its

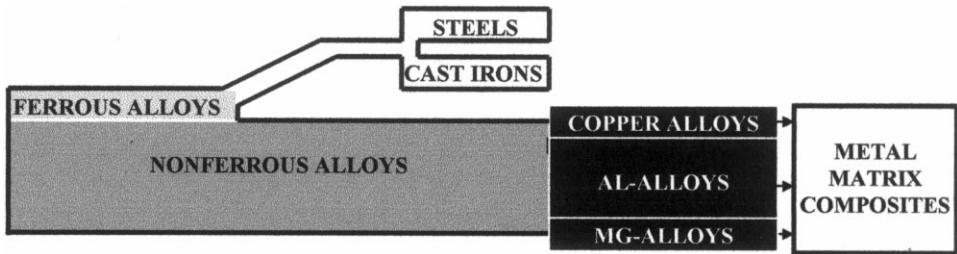


Fig. 5. Schematic diagram to show the alloys used for squeeze casting.

Table 1
Tensile properties of conventional and squeeze cast alloys

Alloy	Yield strength (MPa)	Tensile strength (MPa)	Elongation (%)
AA603-T6 [38]	260	350	≥10
A356-T6 [39]	220	300	≥10
LM24 (SC) [40]	–	210 ± 5	2–2.5
7010 chill cast-T6 [41]	468–488	523–526	4.7–5.9
7010 squeeze cast-T6 (50 MPa) [41]	470–490	550–563	10–14

composition for squeeze casting. The selected alloy was Al–12 wt.% Si–3 wt.% Cu–0.7 wt.% Mg that showed 10–20% improvement in hardness, tensile strength, and elongation over those of gravity cast products. Modification with Sr, Ti, and B of this alloy resulted in 40% increase of elongation, although no significant change in wear resistance was found. Yen and Evans [43] patented an Al casting alloy that consists of, by wt.%: 7.0–13.0 Cu, 0.4–1.2 Mn, 0.21–0.40 V, 0.31–0.70 Zr; impurities are limited to: <0.6% Si, <0.8% Fe, <0.2% Zn, <0.1% Mn, <0.2% Ni, the remainder being essentially Al. The alloy has a tensile strength of ~420 MPa, a yield strength of ~340 MPa, a tensile elongation of ~6%, and a tensile modulus of elasticity of ~80 GPa. A new Mg alloy specifically designed by Chadwick [44] for the squeeze casting process has been tested to show excellent creep and high temperature fatigue properties. The alloy, which was patented in 1987 [45], contained 10–25% Zn, 0.5–5% Cu and 0.25–4% Si. Additions may include up to 1% Ca with a preferred 0.3% Ca and 0.002–0.005% Be.

Squeeze casting process gives new opportunities to fabricate advanced materials, especially in the field of composites. There are a large number of publications devoted to MMCs fabricated by squeeze casting [18,46–52].

Squeeze casting may also be used to fabricate bi-metals where, for instance, cast iron inserts can be incorporated to increase wear resistance in Al alloy components. Applications to date have been wheels, pistons and brake discs.

6. Direction for further development

Fig. 7 reflects the chronology of the developing SC process Direct SC was developed first but is used occasionally to produce simple, symmetrical castings such as pistons, callipers and master cylinders. The use for direct SC is very limited due to the features of the process and the die design: the maximum weight of casting is usually no more than 10 kg. The backward process is more suitable for industrial

needs, because it enables the production of intricately shaped castings such as alloy wheels for the auto industry.

The development of indirect vertical squeeze casting (VSC) machines, essentially extended the application of the squeeze casting process. It allowed the fabrication of complex castings such as alloy wheels with practically no internal defects. Die coat is not required and a metal pressure of up to 100 MPa is applied throughout solidification. The maximum casting size is limited by the machine capacity. However, the drawback of the indirect VSC process is its inability to cast thin sections with a wall thickness of less than 4 mm, due to its low filling velocity and premature solidification of the molten alloy. Furthermore, there are restrictions on the shape complexity of the cast product, which may be due to difficulties experienced with casting ejection.

Fig. 8 illustrates a typical operation of a VSC machine with the following features of: (i) vertical clamping — vertical injection system prevents the inclusion of air and maintains the high temperature of molten metal, and (ii) tilting — a docking short unit mechanism enables the pouring and the clamping at the same time to minimise the cycle time.

Two-stage clamping realises the effective evacuation of air from the cavity through the parting line of die set. A clamping unit achieves quick and shockless clamping. A shot speed stabilising hydraulic circuit keeps the intended casting conditions constant. The technology of vertical squeeze casting with two-plunger system exerting 315–800 tons of clamping force at 50–80 mm s^{−1} provides pore-free aluminium components that are weldable and heat-treatable with physical characteristics approaching those of forgings at 25% less cost [53]. The lower die temperature of VSC, relative to conventional die-casting, produces a faster chill rate with finer grain structure. The VSC castings are more homogeneous than the forgings characterised by grain orientation. The principal applications of vertical shot system technology have been in the automotive industry, business machines and bicycles.

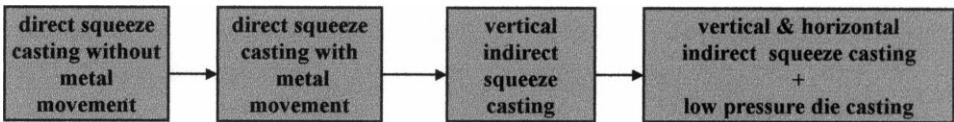


Fig. 7. Development of the squeeze casting process.

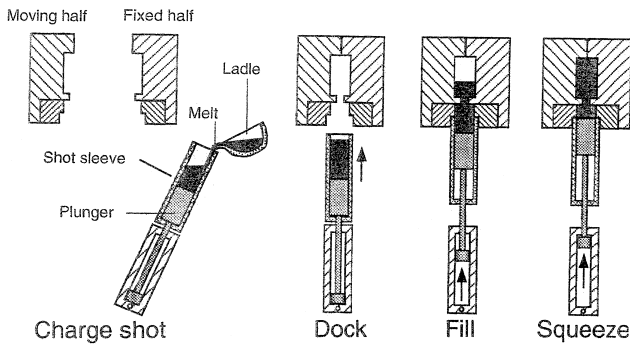


Fig. 8. Schematic diagram to show the operation of a typical vertical injection, horizontally clamped, indirect squeeze casting machine [35].

The next stage in the development of squeeze casting machines is its combination with low pressure die-casting. Toshiba machines has produced a unique die-casting system “LEOMACS” which combines magnetic pump injection for molten metal with indirect metal forging (see Fig. 9), that essentially has extended the possibilities for industrial application. The manufacturer claims that the new casting machine possesses the following features [54]: (i) reduced time from the beginning of the supply of the molten metal to the start of the injection process, resulting in the casting of stronger parts in which the development of non-uniformities during the solidification stage is virtually eliminated, (ii) since the liquid metal is not allowed to come into contact with air, virtually oxide-free products can be cast with ease, (iii) low pressure casting facilitates the use of sand cores, (iv) a high sleeve-filling rate and low filling speeds produce high-quality castings that are free of internal defects, (v) shorter ladling times that result in faster production cycles, (vi) the temperature of the liquid metal can be maintained at a low level since temperature decreases are minimal, and (vii) composite materials can also be cast.

These features allow the production of complex castings with thin walls and internal cavities, i.e., engine frames. Nevertheless, high capital cost is still the main factor, which limits wide industrial application of the process.

The next step in developing squeeze casting machines may lie in their combination with rheocasting where stirring of the melt and its injection into the mould and application of pressure may be achieved in a single unit. It may be possible to install magnetic stirrers around the shot sleeve of an indirect VSC to initiate stirring of the molten metal whilst its temperature is falling. At appropriate timing, when a particular solid percentage is achieved, injection and pressurisation is carried out simultaneously. Such machines are yet to be designed and constructed but the authors envisage that their production will not be in the too distant future. The main advantage of this hybrid metal-forming process is longer die life. Such combination is currently being used to fabricate composite artefacts using rheocasting and squeeze casting as two separate units [52].

The concept of squeeze casting was employed [55,56] to develop the pressure infiltration process. With this process, fibre, whisker or particulate pre-forms are placed in inexpensive throwaway containers, into which the molten matrix material is forced by gas pressure. Because the gas pressure on the preform container is quasi hydrostatic, its strength requirement is minimal; consequently, preform containers, even those with complicated shapes, are easy to prepare. The process is particularly suitable for research and limited production.

Finally, it may be beneficial to compare squeeze casting with other casting processes to highlight the advantages of squeeze casting. This is given in Fig. 10 (from [57]). The overall advantages, based on the cited literature and also on the authors’ own experience, are as follows: superior mechanical properties, fine structure, minimal porosity,

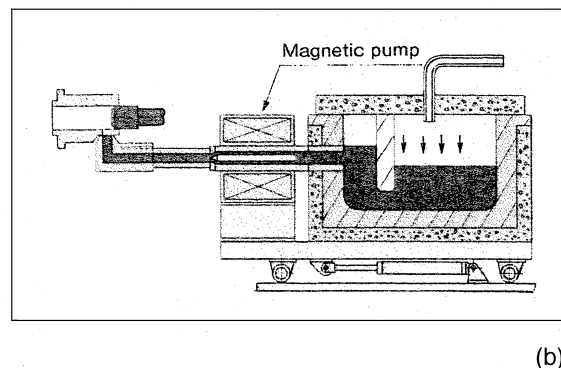
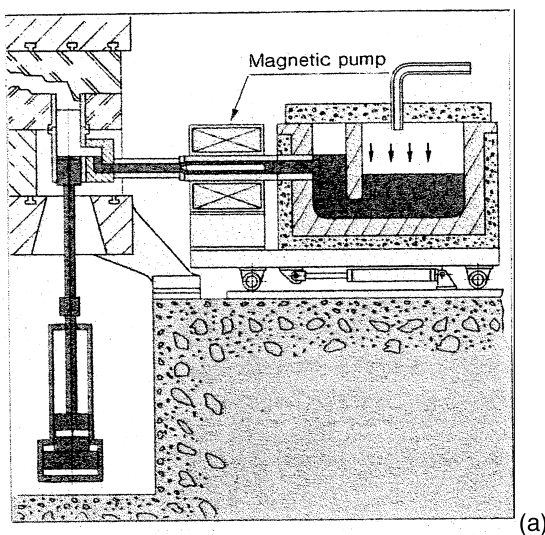


Fig. 9. Schematic diagram of the LEOMACS die-casting system designed by Toshiba Machines, Japan [54]: (a) DXV-vertical die closing and upwards injection, and (b) DXH-horizontal die closing and horizontal injection.

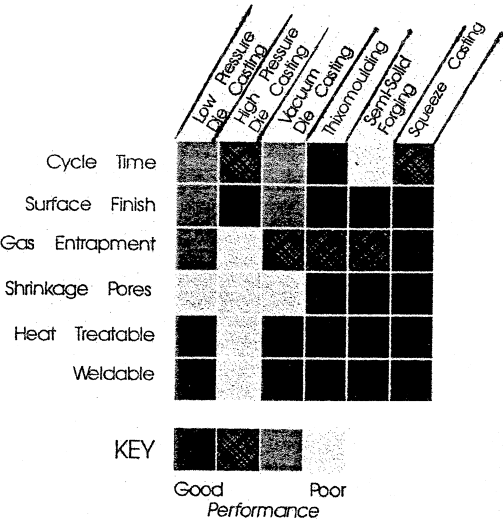


Fig. 10. A graphical representation of casting quality with respect to the casting technique [57].

heat-treatable, weldable, good surface finish, high productivity, applications for composite fabrication, and can cast special alloys.

Among the disadvantages, the following are mostly common listed: high capital cost, shortened die life, limited shape complexity, difficult to produce thin sections, and limited maximum size and weight.

However, as mentioned in this section, most of these limitations are overcome by the advent of more sophisticated machines such as “LEOMACS”.

7. Conclusions

A large number of publications on the squeeze casting process indicate that it is still being developed successfully. The majority of publications are related to aluminium- and magnesium-based alloys and, especially their respective matrix-based composites. Thus it may be true to say that the main tendencies of the development of the squeeze casting process, equipment and alloys, is linked to the fabrication of advanced materials, particularly in the field of Al- and Mg-based alloys, and composites. Some progress is being made in respect of high temperature alloys such as cast-iron but the main emphasis is on light alloys of Al and Mg.

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